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PARAMETER IDENTIFICATION OF LUGRE FRICTION MODEL: EXPERIMENTAL SET-UP DESIGN AND MEASUREMENT

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ABSTRACT

This work proposes a simple but general experimental approach including the rig design and measurement procedure to carry out a wide range of experiments required for identifying parameters for LuGre dynamic friction model. The design choice is based on accuracy of the estimated friction and flexibility in terms of changing contact conditions. The experimental results allow a complete LuGre model, which facilitates, but not limited to, other advanced friction modeling and high performance controller design if needed. In addition, several well-known dynamic friction features (varying break-away force, friction lag and presliding) are successfully demonstrated by our rig, which indicates the adequacy of our approach for capturing highly sophisticated and dynamic friction behavior over a wide range of operating conditions. The proposed set-up and the produced experimental data are believed to greatly facilitate the development of advanced friction compensation and modeling in friction affected mechanisms.

INTRODUCTION

Friction is present in all mechanical systems and generally regarded as a combined result of various factors such as material of frictional surfaces, type of lubricant added into the moving surfaces, thickness of lubricant film, relative velocity of the surfaces in contact and operating temperature. Extremely com-

plicate nature of friction results in the difficulty finding a model that adequately captures all the friction features. Most of the time a viscous friction model is appropriate to interpret the friction behavior due to the fact that the contact surfaces are entirely separated by the lubricant used, such a model is a static model merely taking the velocity as the input. However, more dynamic friction features were found decades ago, and so far there is only a few friction models adequately covering all the dynamic features with the reasonable model complexities which facilitates the model parameter identification and the real time-time computation needs. LuGre friction model [1] has been proved to succeed in accounting for many characteristics of friction, the parameter identification of which is however a demanding task that heavily depends on the resolution of the displacement sensor and the controllability of the experimental rig [2]. Normally, so-called presliding, usually reported a few micrometers in the literatures [3], should be measured for identifying the dynamic parameters for the model, and precise force/velocity controls should be performed to obtain the static parameters. This work focuses on proposing a simple but general experimental approach including the rig design and measurement procedure to carry out a wide range of experiments mentioned above. The experimental results allow a complete LuGre model, which facilitates, but not limited to, other advanced friction modeling and high performance controller design if needed.

A typical experimental rig used to explore the friction features based on translational motion is the friction oscillator [4-6].

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Such a rig normally consists of a vibrating base, several dampers, coil springs and frictional surfaces. The rig is unsealed, which results in easier to alter the loads, materials, and lubricants in contact. However, its complicated structure might hinder the precise friction estimation due to the accumulated error, also none of this type of rigs in the literatures is able to measure the presliding. DC motor is another popular device to study the friction induced by rotary motion [7–10]. However, the frictional surfaces are in sealed environment, which does not allow the factors influencing friction to be easily modified.

In this paper, a simple motor-driven one-dimensional sliding block is proposed and fabricated to carry out the parameter identification for LuGre model. The design choice is based on accuracy of the estimated friction and flexibility in terms of changing contact conditions. The block slides on a metal table, the applied force, displacement and the velocity of which are monitored and controlled by a motor-ballscrew-combined mechanism. In addition, a dedicated capacitive displacement sensor provides extremely high resolution so that displacement of micrometers are measurable using our rig.

The rest of this paper is organized as follows. We first present the design of the mechanical testing structure, sensing system and the friction estimation procedure. Afterwards, the LuGre model will be briefly reviewed, and the parameter identification of which will be discussed and performed by our rig to obtain the nominal parameters. In addition, several well-known dynamic friction features (varying break-away force, friction lag and presliding) are successfully demonstrated by our rig to show the rig's capability to provide accurate dynamic data that facilitates a better understanding and modeling of the selected dynamic features. Finally, the main conclusions are outlined in the last section.

EXPERIMENTAL SET-UP

Design of Experiment Table

The experiment table shown in Fig. 1 mainly consists of a ball screw, a motor and a brass block. The ball screw is coupled with the motor shaft by a beam coupler. One optical encoder is used to provide information about the motion of the shaft. A thin metal plate is bolted to the ball screw nut to transfer the longitudinal motion to the brass block. The load cell, with male thread studs, provides a rigid connection between the thin metal plate and the brass block. The micrometer head is used to calibrate the dedicated high-sensitivity sensor which is required in the identification of parameters in the LuGre friction model. Detailed information on the design, working principle, installation and sensitivity will be addressed later.

The rotary motion of the shaft is transferred to the ball screw shaft via the beam coupler. The resultant rotary motion of the shaft moves the ball screw nut linearly along the longitudinal axis. The high stiffness of the metal plate and the load cell en-

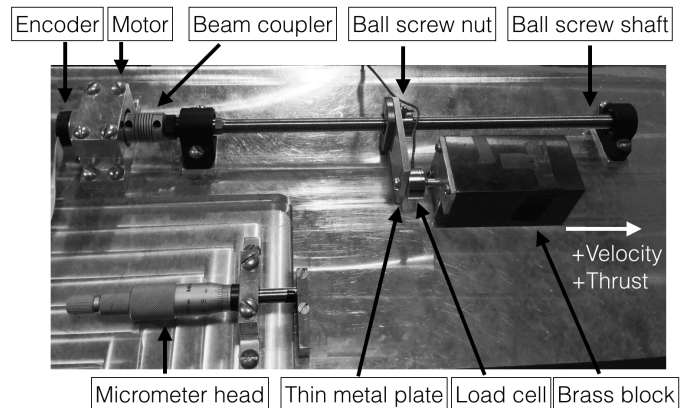


FIGURE 1: PHOTO OF THE SLIDING TABLE.

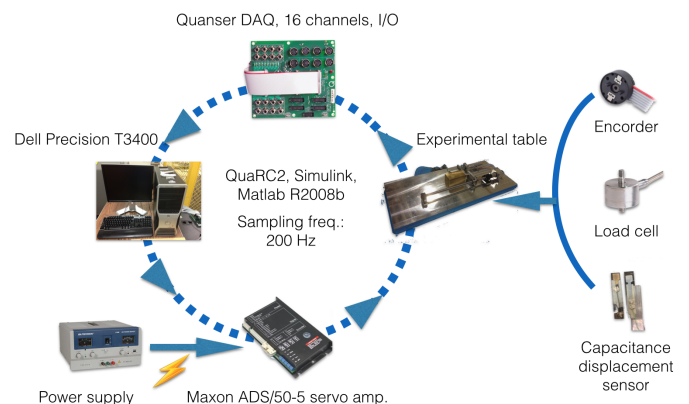


FIGURE 2: THE MAIN CONTROL LOOP, POWER SOURCE AND SENSING SYSTEM.

ures that one dimensional sliding motion is accurately reproduced on the brass block. The load cell is used to continuously measure either the tension or the compression force applied on the brass block. An optical encoder is used to measure the linear displacement of the mass. Such an encoder generates up/down pulses with a resolution of 512 count per revolution. The frictional behavior between the brass block and the metal surface under it is measured and studied in this work. Table 1 summarizes the electrical/mechanical parameters of the developed experiment table.

Figure 2 demonstrates how the experiment table interacts with the PC, servo motor amplifier and the Quanser data acquisition (DAQ). To regulate the controlled output of the brass block, a PID controller is implemented within the Matlab Simulink environment. The PC is manufactured by Dell, it features a dual-core processor with a clock speed of 2.40 GHz and 4 GB of Ram.

TABLE 1: PARAMETERS OF THE MOTOR, BALL SCREW AND LOAD CELL.

Symbol and name	Nominal value	Equipment
m_{motor} , weight of motor	71 [g]	Motor
R , terminal resistance	21.4 [Ω]	
L , terminal inductance	0.953 [mH]	
K_m , torque constant	37.4 [mNm/A]	
K_v , speed constant	255.0 [rpm/V]	
J_m , rotor inertia	4.2 [gcm^2]	
$V_{nominal}$, nominal voltage	36.0 [V]	
P , assigned output power	11.0 [W]	
η_m , max. efficiency at nominal voltage	83 [%]	
Screw shaft diameter	10 [mm]	Ball screw
Lead	1 [mm]	
Thread direction	Right	
Screw thread length	262 [mm]	
Total length	310 [mm]	
Excitation	5 [V]	Load cell
Output	2 [mV/V]	
Capacity	44.5 [N]	
Resolution	0.089 [N]	
Safe overload	66.75 [N]	
Ultimate overload	133.5 [N]	
Weight	1 [kg]	Brass block

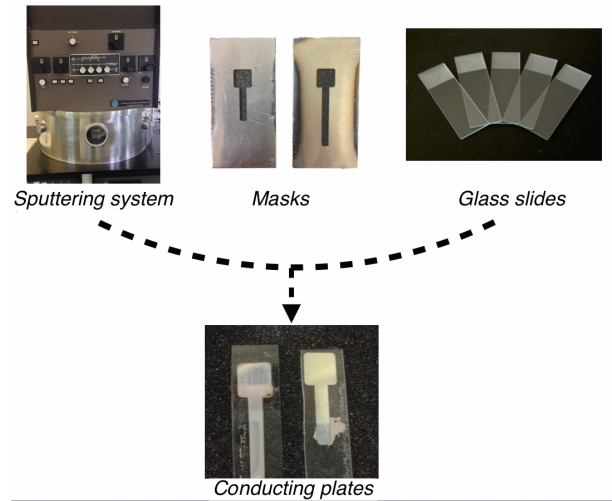


FIGURE 3: TOOLS TO BUILD THE SENSOR.

The amplifier, Maxon ADS/50-5, bridges the controller voltage commands sent from the PC to the motor installed on the experiment table. The controller action ensures that the dynamics of the brass block is under control. Collection of the encoder measurements and the load cell data is performed by the Quanser DAQ. Such measurements will be taken to form the error signal which allows the controller to regulate the output of the brass block. All the realtime measurements available at Quanser board are transmitted into the PC at 200 Hz.

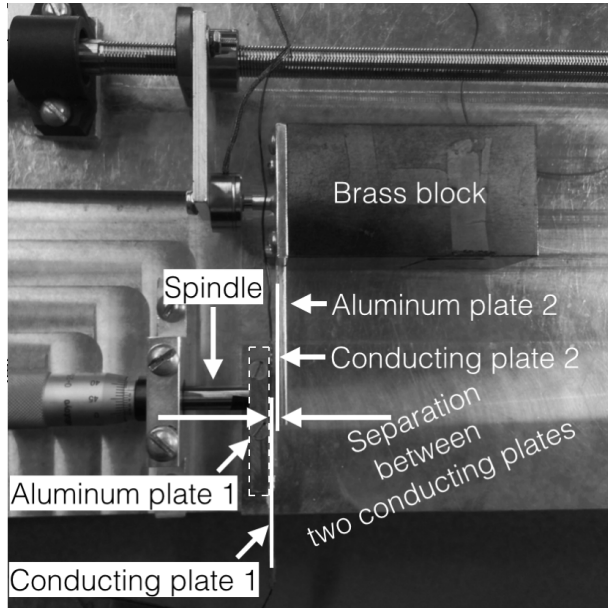
Measurements of the microfine displacement of the block, caused by the applied force lower than the break-away force, has posed a well-known challenge to any encoder-based displacement sensing system. Such a displacement, alternatively named presliding, is essential to the identification process of the LuGre model and is typically on the scale of a few micrometers. To record the presliding of the block, a capacitive displacement sensor featuring high sensitivity was designed and built. The detailed information of the displacement sensor is addressed in the following section.

Design of Capacitive Displacement Sensor

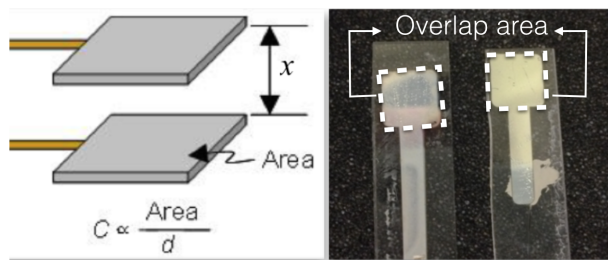
Measuring the aforementioned microfine displacements necessitates a capacitive displacement sensor, which is introduced

in this section. Figure 3 shows all components included in the sensor, which consists of two conducting plates. Each conducting plate is fabricated by depositing a thin layer of aluminum on one piece of glass slide using the sputtering system. The stencil mask helps define the pattern of the conducting plate. Placing two conducting plates in parallel forms a capacitor. The resultant capacitance varies with the separation between two plates if the overlap area and the background capacitance remain unchanged. It is named as a capacitive displacement sensor because of this feature and is widely used in application requiring precise position sensing.

Figure 4a depicts the installation of two conducting plates. Conducting plate 1 is attached on aluminum plate 1 bolted to the spindle. Conducting plate 2 is attached on aluminum plate 2 bolted to the brass block. The two conducting plates are positioned and aligned in a way that its separation gives measurable capacitance, and that the edges of the two squared plates are precisely aligned to form an overlap area. The displacement measured by the sensor consist of two parts. The first part is the true presliding induced by the combined effect of the friction and the applied force, and the second part comes from the elastic deformation of the material of the brass block due to the applied force. To reduce the displacement of the second part, the capacitive displacement sensor is installed at the bottom of the brass block with a clearance of 2mm with respect to the table. This arrangement significantly reduce the second part of the reading from the sensor since the deformation of the block is lower if measured where close to the frictional surface. Figure 4b depicts a typical configuration of the sensor. The capacitance caused by such a parallel-plate capacitor has the form of



(a)



(b)

FIGURE 4: (a) INSTALLATION OF TWO CONDUCTING PLATES. (b) LEFT: SCHEMATIC AND NOTATIONS FOR A PARALLEL-PLATE CAPACITOR. RIGHT: TWO SQUARED CONDUCTING PLATES WITH A SQUARED OVERLAP AREA ARE USED TO FORM THE CAPACITOR.

$$C = \frac{\epsilon A}{x} \quad (1)$$

where ϵ is the permittivity of air, normally $8.854 \times 10^{-12} \text{ F/m}$ is used, A is the overlap area of the two plates, and x is their separation. In the proposed capacitive displacement sensor, the gap x between two conducting plates is designed to be variable. Using the capacitance value in Eqn. (1) measured by high-resolution capacitance sensor, one can convert the variation in capacitance to the variation in the gap by Eqn. (1). The capacitive displacement sensor can provide a resolution of $1 \mu\text{m}$ for the presliding measurement.

The sensitivity, S , of this capacitive displacement sensor is defined as the ratio of the change in capacitance with respect to the displacement as

$$S = \frac{dC}{dx} = -\frac{\epsilon A}{x^2} \quad (2)$$

where the sensitivity increases for low x that consists of the initial separation and the displacement. This favorable feature makes the capacitive displacement sensor popular in the application of precise position sensing.

Friction Profile Estimation

All the trials in this study are performed from rest, i.e., the force applied on the block and the displacement are null before we launch each trial. The following procedure ensures null force is achievable. The brass block is positioned where the ball screw nut has equal travel to both ends of the shaft. This adjustment ensures that the back and forth motion required for some identification processes can be achieved. After the ball screw nut is in position, we manually zero the force applied on the block by rotating the ball screw shaft. The incremental encoder used in this study provides the relative angle between the origin and the current position of the motor shaft. The origin is the position of the motor shaft when the power is turned on. The control tasks are implemented once null force and displacement are secured.

Newton's second law is used to estimate the friction as follows:

$$F_{ext} - \text{friction} = m\ddot{x} \quad (3)$$

where the F_{ext} is the input force, m is the mass of the block, and $\ddot{x}$ indicates the acceleration of the block. Angular displacement, $\Delta\theta$, is converted to the linear displacement using

$$\Delta x = \frac{1}{2\pi p} \Delta\theta \quad (4)$$

where p indicates the pitch of the ball screw, $\Delta\theta$ is the rotational displacement of the lead screw shaft in radians, Δx represents the linear displacement of the brass block. The second derivative of the displacement of the mass with respect to time yields the acceleration in Eqn. (3). The input force sensed by the load cell and the linear displacement is filtered using a second order low-pass filter. Friction estimation at each time instance is achievable using Eqn. (3). So far, a valid procedure is proposed to estimate the friction for conducted trials.

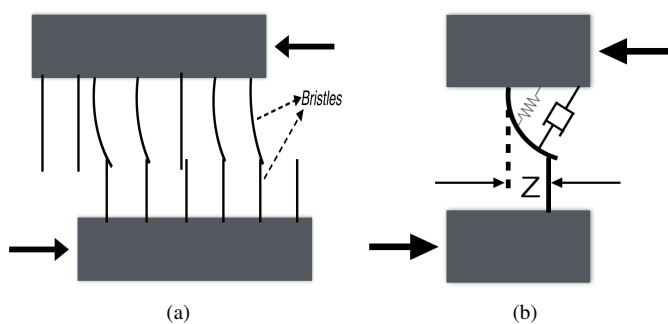


FIGURE 5: (a) CONTACT AREA BETWEEN TWO SOLID BODIES IS THOUGHT OF AS A CONTACT BETWEEN BRISTLES. (b) THE AVERAGE DEFLECTION OF THE BRISTLES BETWEEN THE CONTACT AREA IS REPRESENTED BY A SINGLE BRISTLE WITH THE AVERAGE DEFLECTION z . USING THE SPRING AND THE DAMPER IS ESSENTIAL TO AVOID THE OSCILLATIONS IN THE AVERAGE DEFLECTION.

LUGRE FRICTION MODEL AND ITS PARAMETERS IDENTIFICATION

LuGre Friction Model

LuGre friction model has been used successfully to address many practical control applications due to its success of demonstrating so many aspects of friction. The authors of LuGre model visualize that two rigid bodies are in contact through elastic bristles, which will deflect, giving rise to the friction force when a tangential force is exerted on the mass. While the tangential force is high enough the bristles becomes over-deflected, the stationary mass then becomes a slider. Figure 5 graphically conveys this notion and the friction has the following form

$$F = \sigma_0 z + \sigma_1 \frac{dz}{dt} + \sigma_2 v \quad (5)$$

$$\frac{dz}{dt} = v - \frac{\sigma_0 |v|}{g(v)} z \quad (6)$$

$$g(v) = F_c + (F_s - F_c) e^{-(v/v_s)^2} \quad (7)$$

where F is the friction, σ_0 is the stiffness of the bristles, σ_1 is the damping of the bristles, σ_2 indicates the viscous friction coefficient, v is the relative velocity between two surfaces, z is the average deflection of the bristle, F_c is the Coulomb friction, F_s is the break-away force, and v_s is the Stribeck velocity. All the parameters can be categorized into two groups: static parameters and dynamic parameters. The first group contains F_s , F_c , V_s and σ_2 , and the second group includes σ_0 and σ_1 . The identification for the two groups of parameters will be addressed in the following section.

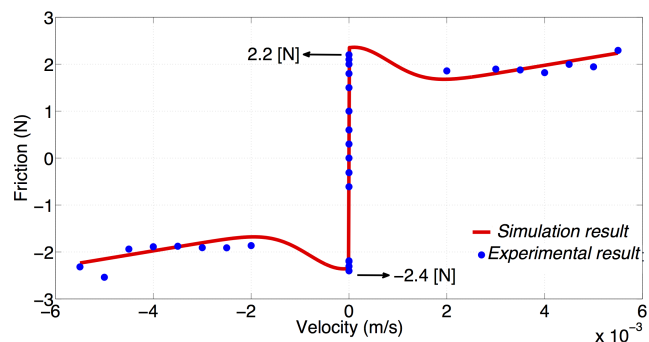


FIGURE 6: FRICTION-VELOCITY MAP: EXPERIMENTAL FINDINGS AND MODEL RESPONSE.

Estimation of Nominal Friction Model Parameters

The static parameters are determined through the friction-velocity map measured during the constant velocity motions. Such a map can be explicitly modeled by the following equation as presented in [1]

$$F = F_c \operatorname{sgn}(v) + (F_s - F_c) e^{-(v/v_s)^2} \operatorname{sgn}(v) + \sigma_2 v \quad (8)$$

To experimentally construct such a map, an experimental test is performed for 15 closed-loop trials at dissimilar desired velocities ranging from $v = \pm 2.0 \text{ mm/s}$ to $\pm 5.5 \text{ mm/s}$ with the equal interval of 0.5 mm/s . A dedicated PID controller regulates the system velocity output. During the steady-state velocity, the friction is therefore equal to the force applied to the block due to the null acceleration. These 15 trials gives the data dots with nonzero velocities plotted on Fig. 6. Afterwards we have to explore the highest load before the dynamic motion of the block is triggered. Thirteen trials are performed by assigning the desired input force to a dedicated PID controller. The maximum loadings before the block motion are experimentally observed as 2.2 and -2.4 N for both directions as can be seen in Fig. 6. At this stage, all the experimental data dots completes the friction-velocity map which can be used to identify the static parameters for our rig.

The curve fitting toolbox in Matlab is used to fit the data dots shown in Fig. 6 to Eqn. (8). The selected method and algorithm are NonlinearLeastSquares and Trust-Region, respectively. Other options are left with the default settings. The identified parameters for positive and negative velocity regions are averaged to obtain the nominal parameters. This post-processing ensures the model predictions to be symmetrical in both regions of velocities. As can be seen in Fig. 6, the model can approximate the experimental results accurately, which makes it practical to numerous engineer applications. All the static parameters are now identified and listed in Tab. 2

The dynamic parameters, namely σ_0 and σ_1 , jointly govern

TABLE 2: NOMINAL STATIC PARAMETERS FOR THE LUGRE MODEL.

Symbol and name	Velocity		Nominal value
	V > 0	V < 0	
F_c , Coulomb friction	1.11	1.458	1.284 [N]
F_s , Static friction	2.3	2.4	2.35 [N]
V_s , Strubeck velocity	0.001943	0.0003365	0.00113975 [m/s]
σ_2 , Viscous friction coefficient	193.4	152.6	173 [N-s/m]

the frictional force and the presliding within the stiction region. It is well-known that these two parameters should be identified by means of the presliding profile caused by step input force with known magnitude [8]. However, measuring the presliding is a challenging task since the high performance displacement sensors are not always available to the researchers. The capacitive displacement sensor presented earlier has been experimentally proved adequate to measure the presliding down to μm level, and the following equation is used to relate the measured presliding, the step input force applied on the block to the parameter σ_0

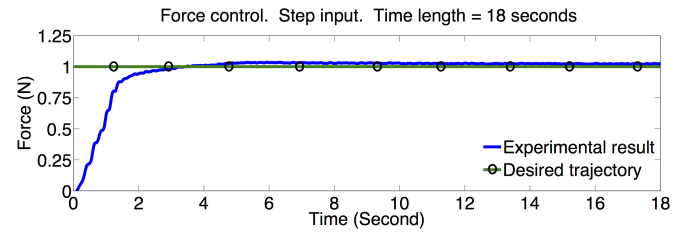
$$\sigma_0 = \frac{Z^T U}{Z^T Z} \quad (9)$$

where Z and U are the vectors containing sample values of the steady-state presliding $z(t)$ and the steady-state step input force magnitude $u(t)$, respectively. An experimental test is performed for 6 closed-loop trials at dissimilar desired step input forces ranging from 0.5N to 1.5N with the equal interval of 0.5N , the details of which are listed in Tab. 3. A dedicated force PID controller regulates the force exerted on the block. Figure 7 shows the input force and the displacement profiles obtained from Trial 4 which is considered as a representative trial.

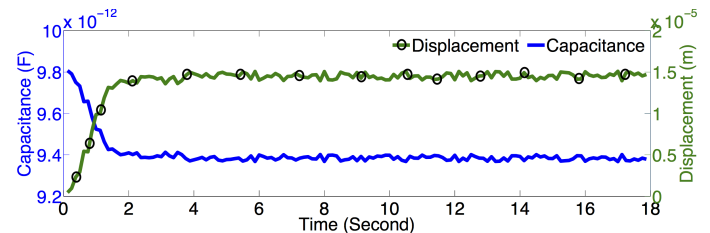
It can be seen in Fig. 7a that the force takes approximately 4 seconds to reach the steady state. The controller gains are carefully tuned to ensure that no aggressive controller action will occur throughout the trials. Such an aggressive action easily leads to the dramatic maximum overshoot in the applied force, which takes the brass block out of the presliding region. Figure 7b shows the variation in the capacitance which decreases due to the increasing separation between two conducting plates. The variation in capacitance is converted to the displacement of the block using the conversion obtained in the calibration process. Such a displacement consists of the presliding and the deformation of the block due to the applied force. In this work the presliding is two orders of magnitude larger than the block deformation as discussed previously, thus the displacement is treated as pure presliding. The presliding in Fig. 7b converges to approximately $15\mu\text{m}$ after the applied force settles to its desired value. With

TABLE 3: TRIALS CONDUCTED FOR IDENTIFICATION PROCESS.

Trial No.	Desired force [N]	Trial No.	Desired force [N]
1	0.5	4	1.0
2	0.5	5	1.5
3	1.0	6	1.5



(a) RESPONSE OF THE APPLIED FORCE TO THE UNIT STEP INPUT.



(b) RESPONSE OF THE CAPACITIVE DISPLACEMENT SENSOR TOGETHER WITH DISPLACEMENT.

FIGURE 7: EXPERIMENTAL RESULTS OBTAINED FROM THE REPRESENTATIVE TRIAL.

TABLE 4: IDENTIFIED PARAMETERS FOR THE LUGRE MODEL.

Trial No.	σ_0 [N/m]	σ_0 nominal	σ_1 nominal
1	65669.888	66967.231	344.56
2	68265.363		
3	67188.616		
4	66851.513		
5	63950.064		
6	69877.942		

measured presliding and applied force, σ_0 can be identified by Eqn. (9).

Table 4 lists all the identified σ_0 values along with their trial numbers. The nominal σ_0 is considered the mean of all the σ_0 values, which is used to determine the nominal σ_1 using the fol-

lowing equation [11]

$$\sigma_1 + \sigma_2 = 2\sqrt{\sigma_0} \quad (10)$$

where the σ_0 and the σ_2 are available at this stage which makes the σ_1 the only unknown in Eqn. (10).

So far all the nominal parameters have been rigorously identified following the procedures provided by the authors of LuGre model. The parameter identification is undoubtedly the most demanding task for the model completion, but it involves a great deal of time and energy and heavily depends on the resolution of the displacement sensor and the controllability of the experimental rig. Based on the experimental results shown in this section, our experimental rig is fully capable of carrying out a wide range of control tasks and capturing all the output profiles required for the identification. A completed LuGre friction model facilitates, but not limited to, investigations of sophisticated frictional behavior, advanced friction modeling and high performance controller design if needed.

DEMONSTRATION OF DYNAMIC FRICTION FEATURES

The desire to understand friction is old but so far there is no any published experimental rig but ours successfully demonstrating all the well-examined static and dynamic friction features on one apparatus. Most of the time, only part of these features are demonstrated and studied in the literatures. This section focuses on the capability of our rig to demonstrate multiple dynamic features. Our rig is exempt from demonstrating the static friction features since they are less challenging than the dynamic ones. Experimental dynamic friction features significantly contribute to advanced friction modeling which is desired to better predict the global, qualitative behavior of a system subject to the friction with a reasonable model complexity. Three dynamic features (varying break-away force, friction lag and presliding) will be demonstrated in the following sections, and a good friction model should be able to simulate which with a good accuracy.

Varying Break-Away Force

Break-away force is the the force to initiate the motion of the object from the rest and is normally greater than the Coulomb friction. It has been experimentally found that break-away force is inversely proportional to the force loading rate rather than a constant. Literatures confirm this feature and provide a thorough analysis of it [12–15]. An experimental test is performed for 20 trials with various force loading rates ranging from 0.08 N/s to 20.3 N/s with unequal intervals, and the corresponding break-away forces will be recorded for the feature demonstration. All the experimental results are plotted on the friction-loading rate map with a base 10 logarithmic scale for the horizontal-axis and

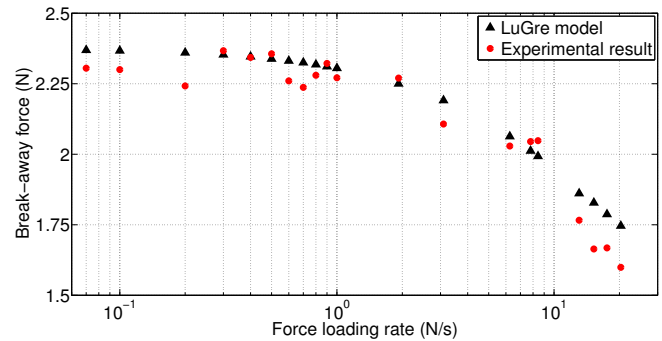


FIGURE 8: RELATION BETWEEN THE BREAK-AWAY FORCE AND THE FORCE LOADING RATE.

a linear scale for the vertical-axis. As shown in Fig. 8, the experimental break-away forces show a dramatic decreasing trend along with the increasing loading rate, which agree qualitatively with the observation in the literatures. Such a good agreement provides us with confidence that the proposed rig is sensitive enough to capture this dynamic frictional behavior.

LuGre model predictions are compared against the experimental break-away forces to examine how good the numerical break-away forces follows the experimental observations. As shown in Fig. 8, the break-away forces reproduced by LuGre model tend to follow the measurements with a maximum deviation of 8% recorded at the loading rate of 20.3 N/s . This good accuracy of the LuGre model implies that the parameters we identified previously using our rig seem to be realistic, and the completed LuGre friction model can express this dynamic feature faithfully and can apply to dissimilar research topics if needed.

Friction Lag

It has been experimentally found that the friction is lower during the deceleration period than the acceleration period when the velocity is continuously varied in one direction within the partial fluid lubrication regime. This phenomenon is named friction lag and attributed to a time delay observed between a changing velocity and the corresponding friction variation. Literatures confirm this feature and provide a thorough analysis of it [4, 6, 16–19]. An experimental test is performed to produce friction lag. The desired sinusoidal velocity trajectory and duration of the control task is presented in Table 5. The friction profile estimated from the task is plotted on the friction-velocity map for the feature demonstration. Figure 9 shows a loop experimentally recorded under the assigned velocity range. The friction is higher for the acceleration period and lower for the deceleration period. The multi-value friction behavior becomes obvious when the velocity is approaching zero, which agree qualitatively

TABLE 5: DESIRED VELOCITY TRAJECTORY OF THE TASK.

Desired velocity trajectory [m/s]	Duration [Seconds]
$1.5 \times 10^{-3} + 1 \times 10^{-3} \sin(3t)$	18.85

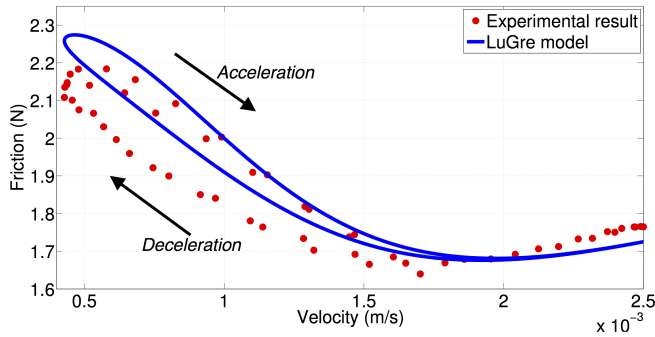


FIGURE 9: FRICTION SHOWS A LOOP PATTERN WHICH CIRCLES CLOCKWISE DURING THE OSCILLATING VELOCITY. TWO SUCCESSIVE CYCLES OF VELOCITY VARIATION ARE PLOTTED FOR THIS PARTICULAR CONDITION.

with the observation in the literatures. This again confirms that the proposed rig is sensitive enough to capture this dynamic frictional behavior.

LuGre model predictions are compared against the experimental friction lag to examine its numerical accuracy. As shown in Fig. 9, the friction lag reproduced by the LuGre model also shows a loop pattern, which follows the experimental observations with a maximum deviation of 7.6% recorded at $V = 5 \times 10^{-4} \text{ m/s}$. As we conclude in the last section, the parameters are realistic and the complete LuGre friction model is adequate to characterize this dynamic feature.

Presliding

Applying an input force lower than the break-away force on the object can produce a microfine displacement between the contact junction. The microfine displacement is usually on the scale of a few micrometers and is alternatively called presliding because such a displacement occurs before the gross sliding. It is well-known that a contact surface behaves like a spring in the presliding region. This is to say, the microfine displacement is approximately a linear function of the applied force. Literatures confirm this feature and provide a thorough analysis of it [3, 20, 21]. An experimental test is performed to produce presliding, the desired input force trajectory with respect to the time of which is presented in Fig. 10. The input force remains at null

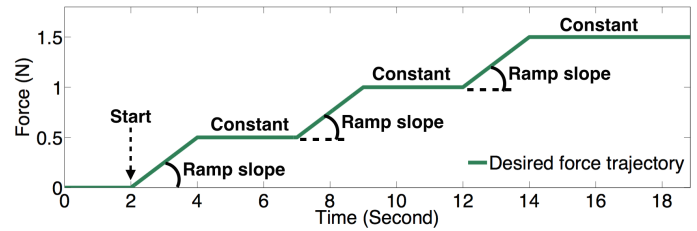


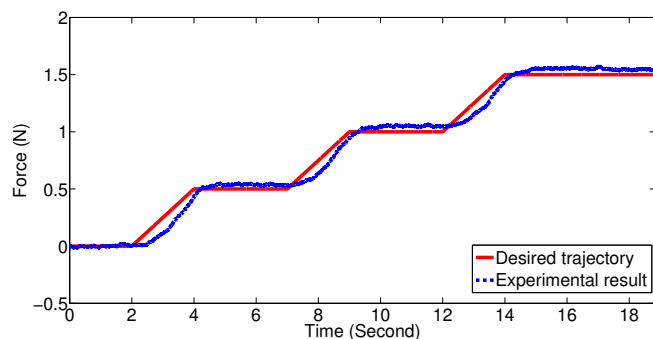
FIGURE 10: DESIRED INPUT FORCE TRAJECTORY USED TO PRODUCE PRESIDING.

for the first 2 seconds and then changes in a linear manner with the ramp slope of 0.25 N/s for another 2 seconds. Afterwards, the desired trajectory will be stabilized at a constant value for 3 seconds and ramped up until it reaches the next constant value as shown in Fig. 10. The time series of the presliding resulting from the test is plotted for the feature demonstration. Figure 11a shows that the applied force follows the desired force with a minor tracking error. The presliding reproduced by the LuGre model is graphically illustrated in Fig. 11b and compared with the experimental presliding. As can be seen in Fig. 11b, the presliding responds quickly to the variation in the applied force without the presence of obvious oscillations, the behavior of which is similar to a critically damped system with the applied unidirectional force, which agree qualitatively with the observation in the literatures. This again confirms that the proposed rig is sensitive enough to capture this dynamic frictional behavior.

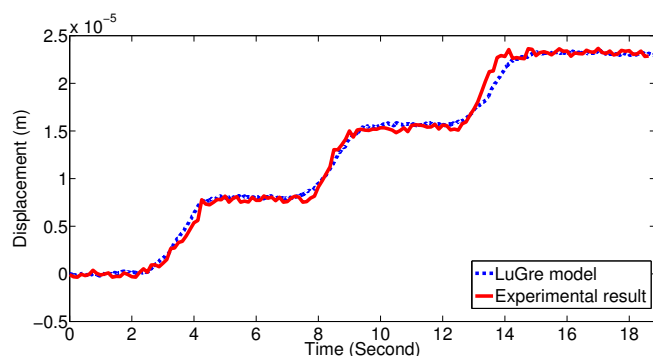
LuGre model predictions are compared against the experimental presliding to examine its numerical accuracy. As shown in Fig. 11b, the LuGre model highly agrees with the experimental results. This high accuracy given by the LuGre on tracking the presliding is attributed to its model structure and the way how two dynamic parameters, σ_0 and σ_1 , are identified. As we conclude in the last section, the parameters are realistic and the complete LuGre friction model is adequate to characterize this dynamic feature.

CONCLUSIONS

Friction is present in all mechanical systems and has an impact on a system's dynamics in a wide range of applications. In some control mechanisms, the frictional forces indeed notably dominate the systems' behavior which hinder the system in meeting the desired characteristics. Therefore, it is nature for researchers to expect a suitable friction model that adequately captures all sophisticated friction features with reasonable model complexities. Such a model can be used to predict/compensate for the friction for the applications of interest. LuGre friction model has been proved to succeed in accounting for many characteristics of friction, the parameter identification of which is however a well-known challenging task.



(a) INPUT FORCE PROFILE.



(b) PRESIDING TOGETHER WITH THE PREDICTION OF THE LUGRE MODEL.

FIGURE 11: PRESIDING MEASUREMENTS ALONG WITH MODEL PREDICTION (RAMP SLOPE=0.25 N/s).

In this work, to facilitate the identification procedure, a dedicated motor-driven one-dimensional sliding block is proposed and fabricated to carry out the identification. The rig design is simple over other existing designs but features the flexibility in terms of changing contact conditions and the sensitive measurement. It has been shown experimentally the rig can be readily used in providing the realistic parameters for the LuGre model. Such a completed LuGre friction model is valuable for the development of advanced friction compensation and the numerical study of friction features. In addition, three well-known dynamic friction features (varying break-away force, friction lag and pre-sliding) are experimentally demonstrated by our rig. These experimental features are important in proposing a new advanced friction model which can fully describe all the friction features with higher accuracy but lower model complexity. In conclusion, the proposed set-up facilitates the development of advanced friction compensation and modeling in friction affected mechanisms, which reduces the undesired effects induced by the friction and improve the system performance (greatest displacement, maximum velocity and smooth motion).

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